Day 5 :- Assemble and Disassemble a PC Practical : Full hands-on session : Assemble and disassemble a desktop PC. Task : Document the process with labeled steps.

Ans. :-

Tasks 1 :- List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Component	Short Description
Motherboard	The main circuit board that connects and allows communication between all computer components.
CPU (Processor)	The "brain" of the computer that performs calculations and executes instructions.
CPU Cooler (Heatsink/Fan)	Keeps the processor cool by removing heat generated during operation.
RAM (Memory)	Temporary storage that holds data and programs currently being used by the CPU.
GPU (Graphics Card)	Processes graphics, videos, and gaming visuals for display on the monitor.
Power Supply Unit (PSU)	Converts AC power from the wall into DC power for computer components.
HDD (Hard Disk Drive)	A storage device that stores operating systems, files, and applications using magnetic disks.
SSD (Solid State Drive)	A faster storage device that uses flash memory instead of moving parts.
Optical Drive (if installed)	Reads and writes CDs, DVDs, or Blu-ray discs.
Power Cables	Supply electrical power from the PSU to all internal components.
Case Fans	Improve airflow inside the cabinet and help keep components cool
Expansion Cards	Additional cards such as Wi-Fi, sound, or capture cards that add extra features.
CMOS Battery	

	Maintains BIOS settings and system clock when the PC is powered off.
SATA Cables	Connect storage devices like HDDs and SSDs to the motherboard.

**Tasks 2 :- Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.**

Desktop PC Disassembly Steps :-

Step 1: Unplug the Power :-

Caption: The computer was powered off and disconnected from electricity to ensure safe disassembly.

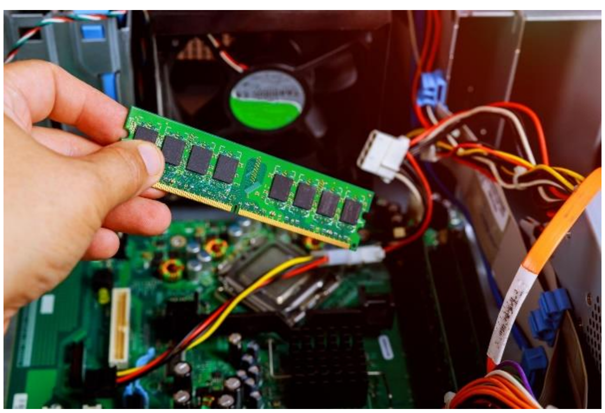
Step 2: Remove the Side Panel :-

Caption: The side panel was removed to gain access to the internal components of the PC.

Step 3: Disconnect Internal Cables :-

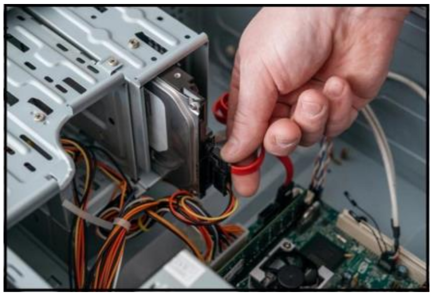
Caption: All power and data cables were carefully disconnected from the motherboard and storage devices.

Step 4: Remove the RAM Module :-



Caption: The RAM module was removed from the DIMM slot after releasing the locking clips.

Step 5: Remove HDD/SSD :-

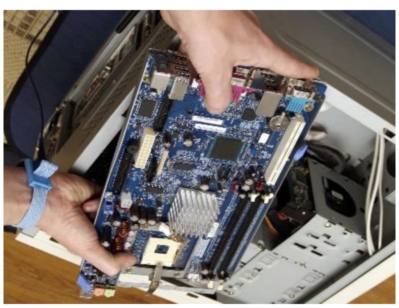


Caption: The storage drive was disconnected and removed from the computer case.

Step 6: Remove the Graphics Card (GPU) :-

Caption: The graphics card was removed from the PCIe slot after disconnecting its power cable.

Step 7: Remove the Motherboard :-



Caption: The motherboard was removed from the cabinet after all connections and screws were detached.

Step 8: Final Inspection :-

Caption: The desktop PC was fully disassembled and all components were inspected for safe handling.

Tasks 3 :- Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters

Step 1: Install the Motherboard :-

The motherboard was carefully placed inside the cabinet and secured with screws. This step is done first because all major components connect to the motherboard.

Step 2: Connect Power Supply Cables :-

The 24-pin ATX power cable and CPU power cable were connected to the motherboard. These connections are required to provide power to the system.

Step 3: Install the RAM Module :-

The RAM module was inserted into the DIMM slot until the retaining clips locked into place. RAM must be installed before the system can boot properly.

Step 4: Install HDD/SSD :-

The storage drive was mounted in its drive bay and secured with screws. This step is necessary because the operating system and files are stored on the drive.

Step 5: Connect SATA Data and Power Cables :-

SATA data cables were connected between the motherboard and storage drive, and SATA power cables were connected from the power supply. These connections allow the drive to communicate with the system and receive power.

Step 6: Install the Graphics Card (GPU) :-

The graphics card was inserted into the PCIe x16 slot and secured with screws. Any required PCIe power connectors were attached. The GPU is needed for video output and graphics processing.

Step 7: Connect Front Panel and Case Fan Cables :-

Front-panel connectors (power switch, reset switch, power LED, HDD LED) and case fan cables were connected to the motherboard. These connections enable system controls and cooling.

Step 8: Verify All Connections :-

All cables, screws, RAM modules, storage devices, and expansion cards were checked to ensure they were properly installed and secure.

Step 9: Reattach the Side Panel :-

The side panel was placed back onto the cabinet and secured with screws to protect the internal components.

Step 10: Power On and Test :-

The power cable was connected and the computer was powered on. The system booted successfully, confirming that all components were installed correctly.

Why the Order Matters :-

1. The motherboard must be installed first because all other components connect to it.
2. Power connections are required before testing components.
3. RAM and storage must be installed before the operating system can load.
4. The GPU is installed after the motherboard to ensure proper seating in the PCIe slot.
5. Final cable checks help prevent boot failures and hardware damage.

Tasks 4 :- Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

PC Assembly/Disassembly Process :-

SAFETY PRECAUTION	WHY IT IS IMPORTANT
Turn off the computer completely	Prevents electrical shock and component damage.
Unplug the power cable	Ensures no electricity is supplied to the system.
Switch off the PSU (Power Supply Unit)	Provides an additional layer of electrical safety.

Press the power button after unplugging	Discharges any remaining electricity in the system.
Use an anti-static wrist strap	Prevents static electricity from damaging sensitive components.
Work on a clean, dry surface	Reduces the risk of electrical hazards and lost parts.
Avoid touching circuit contacts	Prevents damage from static discharge and contamination.
Handle components by the edges	Protects delicate chips and connectors from damage.
Keep liquids away from the workspace	Prevents short circuits and hardware damage.
Store screws and small parts safely	Prevents loss of important components during reassembly.

Tools Required :-

TOOL	WHY IT IS IMPORTANT
Phillips-head screwdriver	Used to remove and install most PC screws.
Anti-static wrist strap	Protects components from electrostatic discharge (ESD).
Small container for screws	Keeps screws organized and prevents loss.
Flashlight or LED light	Helps view small connectors and components inside the case.
Cable ties (Zip ties)	Used for cable management and improved airflow.

Soft cleaning brush	Removes dust from components safely.
Compressed air can	Cleans dust from fans, heatsinks, and other parts.
Thermal paste (if removing CPU cooler)	Ensures proper heat transfer between CPU and cooler.
Microfiber cloth	Cleans surfaces without scratching components.